Anonymous supplement: rational VP Heun certificate

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This supplement records the integer comparison used in the workshop paper. It contains no author names, emails, or public repository URLs.

The four-step VP Heun product is an element $P(\pi, \sqrt{2})$ of $\mathbb{Q}[\pi, \sqrt{2}]$ with twelve monomials, all coefficients nonnegative. Substituting $\pi < 355/113$ and $\sqrt{2} < 99/70$ is therefore valid and yields

$$P\Big(\frac{355}{113}, \frac{99}{70}\Big) = \frac{19211335367141247013930 3}{10275342787993970881331 2} < \frac{187}{100},$$

equivalently

$$1921133536714124701393030 0 < 1921489101354872554808934 4.$$

Hence $r_{\mathrm{VP}} < 187/100$ and $W_2^{\mathrm{VP}} > 13/100$. The bound $\sqrt{2} < 99/70$ is $99^2 - 2 \cdot 70^2 = 1$. The bound $\pi < 355/113$ follows from Machin's formula with Leibniz remainders (seven Taylor terms for $\arctan(1/5)$, four for $\arctan(1/239)$). An independent `mpmath` 1.3.0 interval enclosure is a cross-check, not the proof.